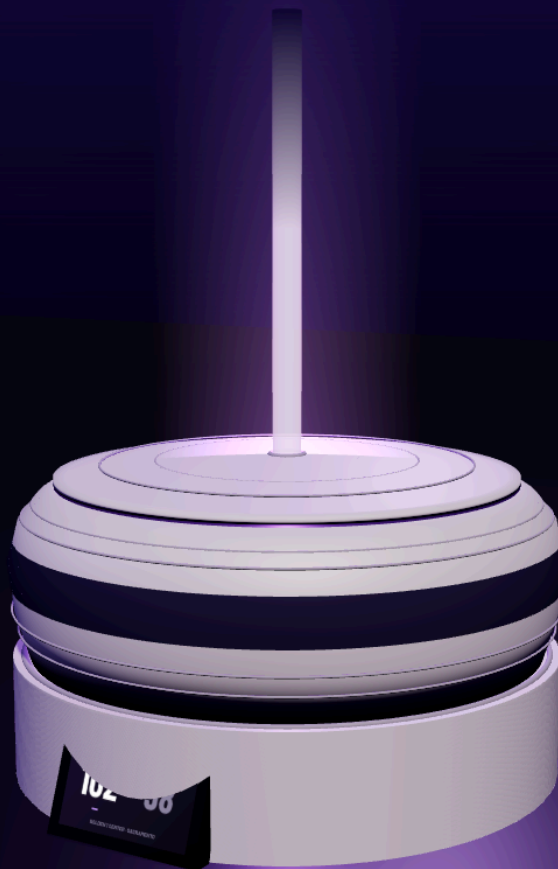




LIGHT THE BEAM

Assembly and setup guide. About 45 minutes once the parts are printed. No soldering, no software to install.

[READ ALL STEPS BEFORE YOU START](#)

What's in the box

A

Arena base, 220 mm
silver PLA · from Printables

B

Arena lid
silver PLA · 12.4 mm beam hole

C

Plinth
silver PLA · screen window at front

D

Plinth bottom lid
silver PLA · press fit

E

Beam tube, 250 mm
white PLA · single wall

F

Strip spine
white PLA · 10 × 262 mm bar

G

Tube socket
white PLA · 30 mm puck

H

Display board
ESP32-2432S028R with USB
cable and pigtail leads

I

LED strip, 1 m
WS2812B 5 V, 60 LEDs/m · you'll
use 15 LEDs

J

3 × lever connectors
WAGO 221-413

K

USB power adapter
5 V, 2 A or more

—

Optional
USB-C panel-mount jack ·
74AHCT125 level shifter

You will need

- ✓ Scissors or flush cutters (for the LED strip)
- ✓ Hot glue gun, or a few dots of super glue
- ✓ A phone, for the one-time WiFi setup
- ✓ Optional: small flat screwdriver for the lever nuts

Printing the parts yourself? Print settings and times are on page 9. Print the plinth's front test slice first and check that the display board fits before committing to the full plinth.

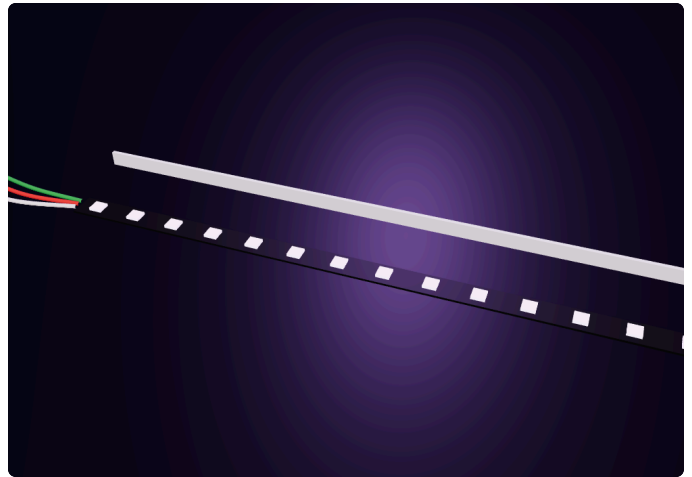
The beam

1

Cut the strip to 15 LEDs

1. Unroll the LED strip. Starting from the end with the **wires attached**, count 15 LEDs.
2. Cut across the copper pads **after** the 15th LED, on the printed cut line.
3. Note the arrow printed on the strip: it points away from the wired end. Data flows that way.

Keep the wired end. The three leads (red 5V, green data, white or black ground) are what you'll connect in step 6.

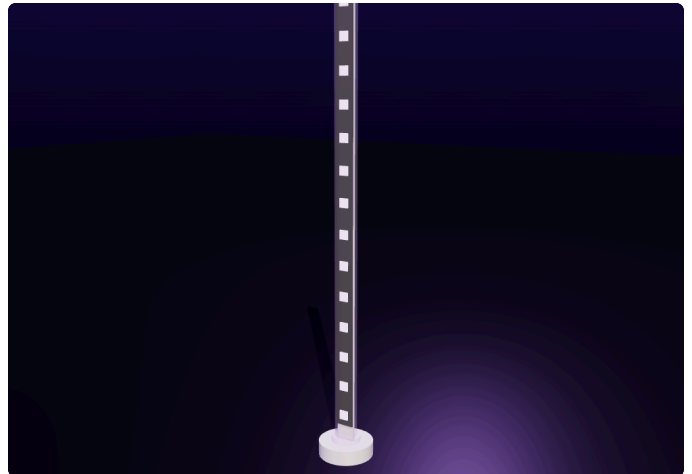


2

Stick the strip to the spine, slide into the tube

1. Peel the strip's backing and press it onto the flat face of spine **F**, wired end at the notched bottom.
2. Feed the leads through the notch, then slide the spine into tube **E** from the open bottom, LEDs facing the tube wall.
3. Push until the spine's feet stop against the tube's foot flange.

Tight fit? The tube is one thin wall; flex it gently rather than forcing. A dab of hot glue at the foot keeps the spine from sliding out.



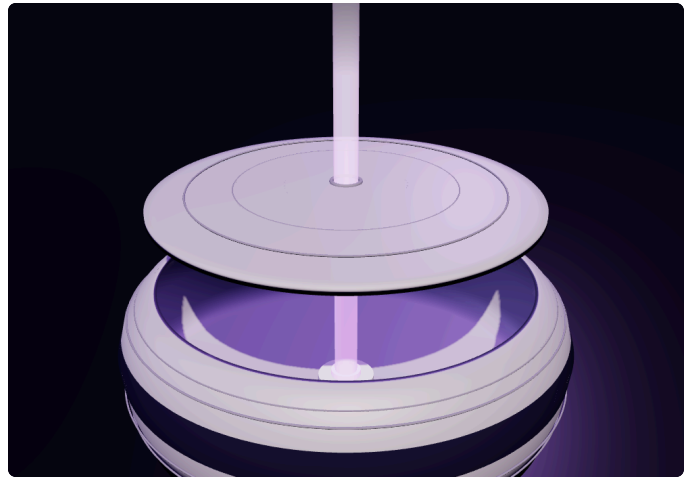
The arena

3

Mount the socket and the beam

1. Find the centre of the arena base **A** floor. Glue socket **G** there, slot facing the arena's wire hole at the back.
2. Pass the three strip leads through the socket's slot and out the arena's wire hole.
3. Drop the tube's foot into the socket. It should stand straight; if it leans, the glue hasn't set.

Wait for the glue. The tube is 250 mm tall; a socket that shifts while curing gives a permanently leaning beam.

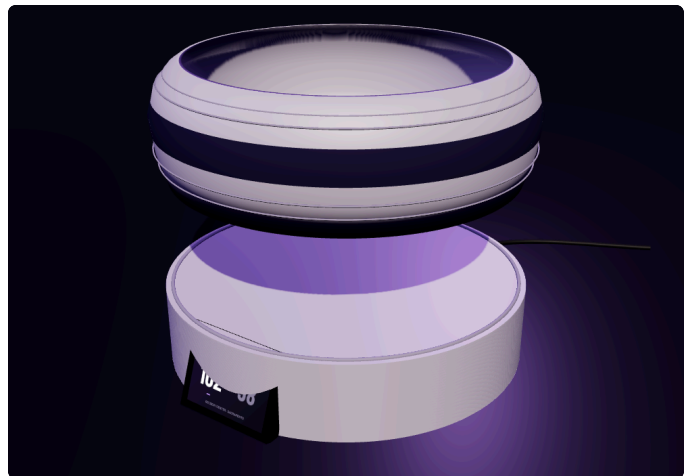


4

Fit the lid

1. Lower lid **B** over the tube; the 12.4 mm hole should clear the tube with a hair of play.
2. Seat the lid on the arena rim. It lifts off again any time you need to reach inside.

Cut your own hole? If you printed the no-hole lid, add a 12.4 mm negative cylinder in Bambu Studio, centred, before slicing.



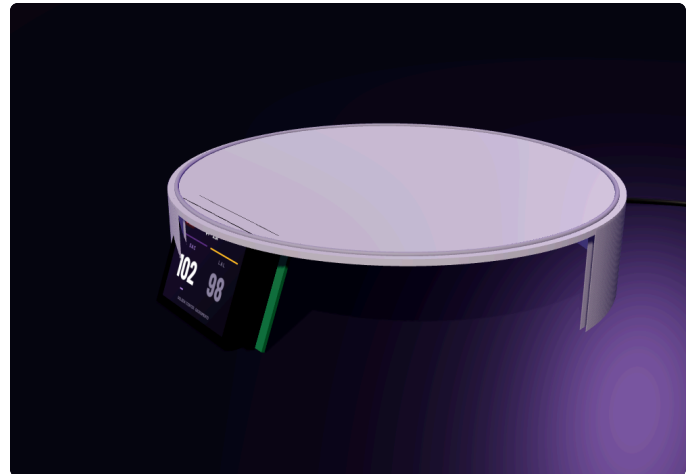
The plinth and the wiring

5

Seat the display board

1. Turn plinth **C** upside down. Behind the front window is a tilted pocket.
2. Slide board **H** into the pocket, **screen facing the window**, USB connector toward the bottom edge.
3. Push until the glass rests against the window lip.
From the front, the picture area should be centred in the window.

Off-centre? The window position is a parameter in the plinth model (`screen_ofs_x`). Adjust and reprint the test slice rather than the whole plinth.



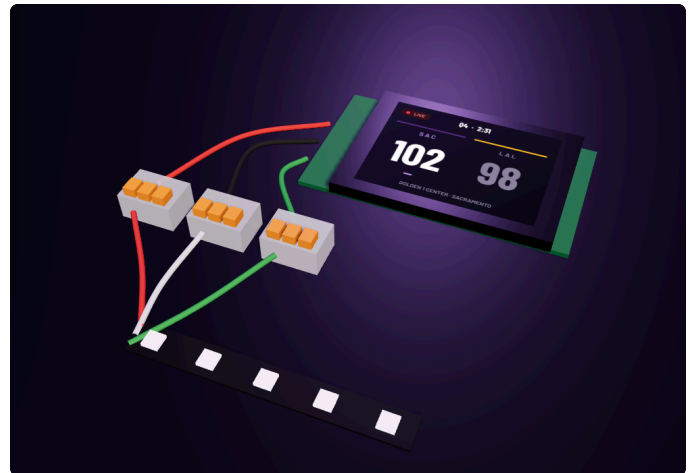
6

Connect the strip: three lever nuts

Each connector **J** joins one board lead to one strip lead. Lift the orange lever, push the stripped wire in to the stop, close the lever. Fold the board's thin pigtail wire double so it grips.

BOARD (P1 / CN1 PIGTAILS)	STRIP LEAD	CARRIES
■ VIN	■ Red	5 V power
■ GND	■ White (or black)	Ground
■ I022	■ Green	Data

Check before power. Red to VIN, green to I022. Swapping 5 V and data can damage the first LED.

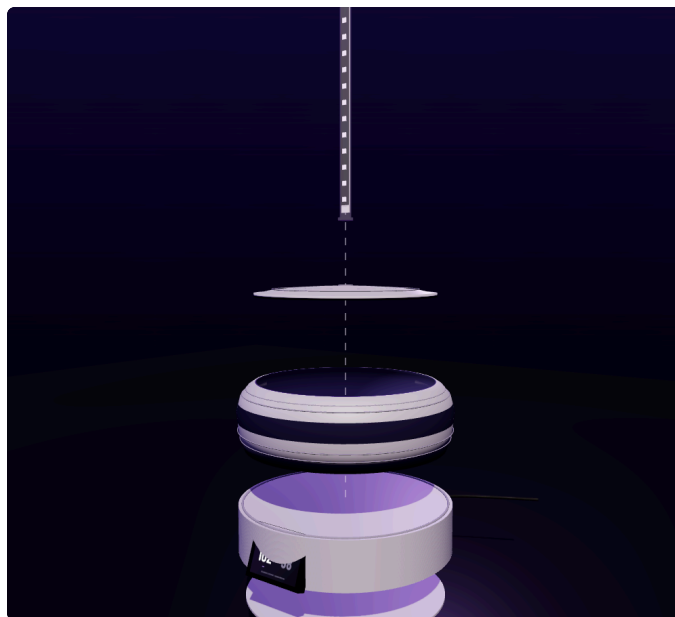


Close it up and power on

7

Route the cable, close the plinth

1. Tuck the three connectors into the plinth. Route the board's USB cable out the slot at the back (or plug it into the panel jack if you fitted one).
2. Press bottom lid **D** into the plinth. The bump on the lid presses the board up against the window.
3. Set the arena onto the plinth; it drops into the shallow recess. Feed the strip leads down the wire hole first.



8

Power on and join WiFi

1. Plug the USB cable into adapter **K**. The screen shows the Kings logo, then **WIFI SETUP**.
2. On your phone, open WiFi settings and join the network **KingsBeam-Setup**.
3. A setup page opens by itself (if not, browse to `192.168.4.1`). Choose your home WiFi, enter its password, tap Save.
4. Within a few seconds the screen changes to **NEXT GAME**. The board remembers your WiFi from now on.

Test the beam now. Tap the screen once: the beam should rise purple from the bottom. Tap again to turn it off.



Using it

What the screen shows

NEXT GAME	Opponent, home or away, tipoff time, countdown. Updates every 10 minutes; every 30 seconds in the last 20 minutes before tipoff.
LIVE	Score, quarter and clock, refreshed every 20 seconds. The beam pulses softly.
FINAL	Kings win: the beam rises and holds, and the screen says so. Loss: the score and "next time".
OFFLINE	A small badge in the corner. The last good screen stays up; it retries every minute.

Touch

TAP	Beam on / off. Works in any state.
HOLD 1 S	Cycle beam brightness: low, medium, full.

Automatic behaviour

AFTER A WIN	The beam stays lit for up to 12 hours, until you tap it off, or until the next game appears.
AT NIGHT	Screen dims from 11 pm to 7 am, and whenever the room is dark.
POWER LOSS	Reboots into the current state; WiFi is remembered.

Care

Dust with a dry cloth. Keep away from windows in direct sun: PLA softens above about 55 °C. The lid lifts off to reach the beam; the bottom lid pulls off to reach the board.

Moving to another WiFi

If the board can't find the saved network within 20 seconds of booting, it opens **KingsBeam-Setup** again. Join it and repeat step 8.

Troubleshooting

SYMPTOM	TRY
Screen dark, no logo	Use a 2 A adapter, not a laptop port. Try the other USB socket on the board if it has two.
Colors inverted or washed out	You have the other board variant. Rebuild the firmware with the ST7789 flags noted in <code>firmware/platformio.ini</code> .
Picture upside down	Change <code>LV_DISPLAY_ROTATION_90</code> to <code>_270</code> in <code>firmware/src/ui.cpp</code> , rebuild.
Beam never lights, even on tap	Check red → VIN and green → I022. Confirm the strip's arrow points away from the wired end.
First LED flickers or wrong color	Shorten the data lead, or add the 74AHCT125 level shifter between I022 and the strip.
Beam lights but dim	Hold the screen for a second to step brightness. Or you're on a weak charger.
"OFFLINE" badge won't clear	ESPN is unreachable. It retries every minute. If it persists for hours, reboot; if it still persists, the firmware's user agent may need updating.
Wrong time on the countdown	The board syncs time from the internet on boot. Give it a minute after connecting.
Taps not registering	Press firmly with a fingertip; the panel is resistive. Threshold is <code>Z_THRESHOLD</code> in <code>firmware/src/touch.cpp</code> .

Printing the parts

PART	FILE	FILAMENT	SETTINGS	TIME
A · Arena base	Printables, 220 mm base	silver PLA	0.2 mm · 3 walls · 15 % · no supports	~9 h
B · Arena lid	Printables, no-hole lid + 12.4 mm cut	silver PLA	0.15 mm	~2 h
C · Plinth	plinth.stl (test plinth_test.stl first)	silver PLA	0.2 mm · upside down · 15 % gyroid	~5 h
D · Bottom lid	plinth_lid.stl	silver PLA	0.2 mm	~40 min
E · Beam tube	beam_tube.stl	white PLA	vase mode · 0.4 nozzle · 2 bottom layers	~1 h
F · Spine, G · Socket	beam_spine.stl , beam_socket.stl	white PLA	0.2 mm	~20 min

Specifications

FOOTPRINT	230 mm diameter
HEIGHT	≈ 385 mm to the beam tip
POWER	5 V USB, 1 A peak
NETWORK	2.4 GHz WiFi
DATA	ESPN public scores, no account

Safety

Low-voltage USB only. Don't power from anything but a 5 V adapter. Indoor use. The beam tube is thin-walled: don't let small children swing it. Unplug before opening the plinth.